```sql
--NIVEL 3--

--EJERCICIO 1--

SELECT
    cat.categoria_id,
    cat.nombre AS categoria,
    SUM(dp.cantidad) AS unidades_vendidas,
    ROUND(
        SUM(dp.cantidad * (dp.precio_unitario - pr.precio_compra)) /
        SUM(dp.cantidad * pr.precio_compra) * 100
    , 2) AS margen_ponderado_pct
FROM detalle_pedidos dp
INNER JOIN productos pr ON dp.producto_id = pr.producto_id
INNER JOIN categorias cat ON pr.categoria_id = cat.categoria_id
INNER JOIN pedidos pe ON dp.pedido_id = pe.pedido_id
WHERE pe.estado <> 'cancelado'
GROUP BY cat.categoria_id, cat.nombre
ORDER BY margen_ponderado_pct DESC;
```

categorias 1

Enter a SQL expression to filter results (use Ctrl+Space)

Grilla | Texto

	categoria_id	categoria	unidades_vendidas	margen_ponderado_pct
1	4	Televisores	1.313	44,63
2	6	Gaming	1.271	41,05
3	14	Energía	1.975	39,2
4	12	Componentes PC	1.763	39,06
5	11	Impresoras	2.002	37,78
6	3	Tablets	1.731	37,53
7	2	Laptops	1.633	37,02
8	9	Hogar Inteligente	1.102	36
9	7	Cámaras	2.097	35,97
10	8	Wearables	1.301	35,95
11	13	Redes	1.535	35,65
12	15	Software	1.937	34,15
13	10	Accesorios	1.233	34,06
14	1	Smartphones	2.062	33,76
15	5	Audio	2.006	32,6

```sql
--EJERCICIO 2--

SELECT DISTINCT
    c.cliente_id,
    c.nombre,
    c.apellido,
    c.email
FROM clientes c
INNER JOIN pedidos p2023 ON c.cliente_id = p2023.cliente_id
WHERE p2023.fecha_pedido >= '2023-01-01' AND p2023.fecha_pedido < '2024-01-01'
  AND p2023.estado <> 'cancelado'
  AND NOT EXISTS (
      SELECT 1
      FROM pedidos p2024
      WHERE p2024.cliente_id = c.cliente_id
        AND p2024.fecha_pedido >= '2024-01-01' AND p2024.fecha_pedido < '2025-01-01'
        AND p2024.estado <> 'cancelado'
  )
ORDER BY c.cliente_id;
```

clientes 1

Enter a SQL expression to filter results (use Ctrl+Space)

Grilla | Texto | Record

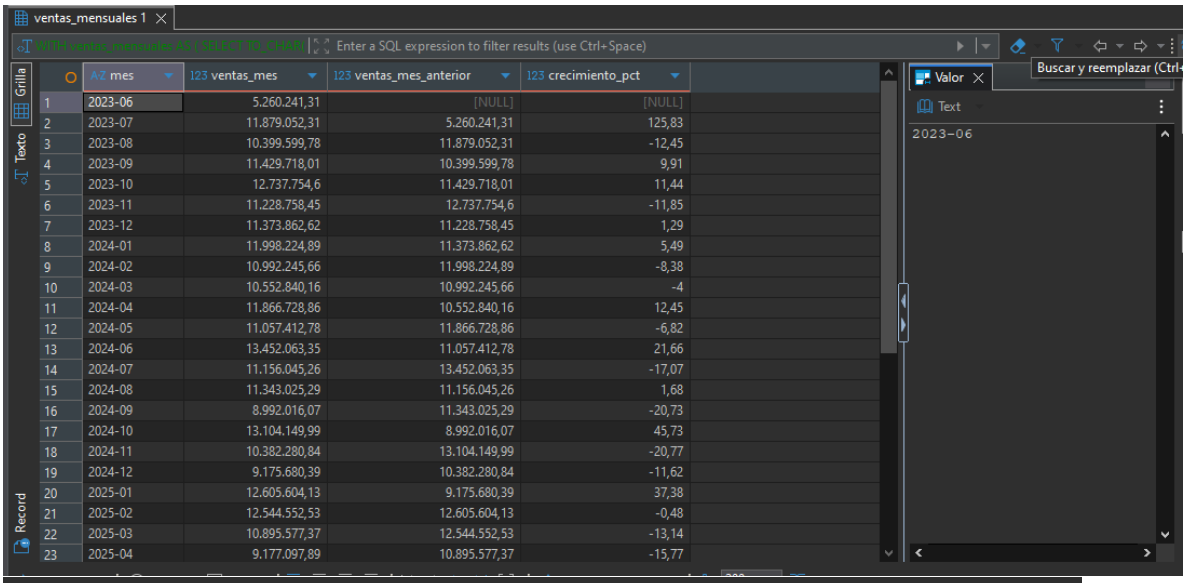
	cliente_id	nombre	apellido	email
1	3	Claudia	Vega Morales	claudia3@hotmail.com
2	14	Patricia	Vega Vega	patricia14@gmail.com
3	19	Sandra	Torres Pérez	sandra19@hotmail.com
4	23	Pablo	Ortega García	pablo23@yahoo.com
5	26	Alejandro	López Gómez	alejandro26@gmail.com
6	34	Manuel	Hernández Reyes	manuel34@hotmail.com
7	44	Mónica	Vargas González	mónica44@gmail.com
8	50	Juan	González Reyes	juan50@gmail.com
9	70	Manuel	Mendoza Vargas	manuel70@gmail.com
10	81	Fernando	Reyes Martínez	fernando81@yahoo.com
11	86	Daniela	Cruz Cruz	daniela86@gmail.com
12	88	Isabel	Núñez Gutiérrez	isabel88@gmail.com
13	94	Manuel	Ortega González	manuel94@hotmail.com
14	95	Andrés	Herrera Jiménez	andrés95@gmail.com
15	104	Ana	Ramírez Rodríguez	ana104@yahoo.com
16	123	Ana	Reyes Martínez	ana123@hotmail.com
17	132	Rosa	García Jiménez	rosa132@gmail.com
18	133	María	Jiménez Castro	maría133@yahoo.com
19	134	Alberto	García Sánchez	alberto134@gmail.com
20	139	Juan	Martínez Castillo	juan139@gmail.com
21	140	Adriana	Hernández Martínez	adriana140@gmail.com
22	158	Carlos	Jiménez Gutiérrez	carlos158@gmail.com
23	159	Sergio	Ortega Sánchez	sergio159@yahoo.com

Valor

3

```sql
--EJERCICIO 3--

WITH ventas_mensuales AS (
    SELECT
        TO_CHAR(fecha_pedido, 'YYYY-MM') AS mes,
        SUM(total) AS ventas_mes
    FROM pedidos
    WHERE estado <> 'cancelado'
    GROUP BY mes
)
SELECT
    mes,
    ventas_mes,
    LAG(ventas_mes) OVER (ORDER BY mes) AS ventas_mes_anterior,
    ROUND(
        ((ventas_mes - LAG(ventas_mes) OVER (ORDER BY mes)) /
        LAG(ventas_mes) OVER (ORDER BY mes)) * 100
    , 2) AS crecimiento_pct
FROM ventas_mensuales
ORDER BY mes;
```



	mes	ventas_mes	ventas_mes_anterior	crecimiento_pct
1	2023-06	5.260.241,31	[NULL]	[NULL]
2	2023-07	11.879.052,31	5.260.241,31	125,83
3	2023-08	10.399.599,78	11.879.052,31	-12,45
4	2023-09	11.429.718,01	10.399.599,78	9,91
5	2023-10	12.737.754,6	11.429.718,01	11,44
6	2023-11	11.228.758,45	12.737.754,6	-11,85
7	2023-12	11.373.862,62	11.228.758,45	1,29
8	2024-01	11.998.224,89	11.373.862,62	5,49
9	2024-02	10.992.245,66	11.998.224,89	-8,38
10	2024-03	10.552.840,16	10.992.245,66	-4
11	2024-04	11.866.728,86	10.552.840,16	12,45
12	2024-05	11.057.412,78	11.866.728,86	-6,82
13	2024-06	13.452.063,35	11.057.412,78	21,66
14	2024-07	11.156.045,26	13.452.063,35	-17,07
15	2024-08	11.343.025,29	11.156.045,26	1,68
16	2024-09	8.992.016,07	11.343.025,29	-20,73
17	2024-10	13.104.149,99	8.992.016,07	45,73
18	2024-11	10.382.280,84	13.104.149,99	-20,77
19	2024-12	9.175.680,39	10.382.280,84	-11,62
20	2025-01	12.605.604,13	9.175.680,39	37,38
21	2025-02	12.544.552,53	12.605.604,13	-0,48
22	2025-03	10.895.577,37	12.544.552,53	-13,14
23	2025-04	9.177.097,89	10.895.577,37	-15,77

```sql
--EJERCICIO 4--

WITH ventas_empleados AS (
    SELECT
        d.region_id,
        e.empleado_id,
        e.nombre || ' ' || e.apellido AS vendedor,
        SUM(p.total) AS total_ventas
    FROM pedidos p
    INNER JOIN empleados e ON p.empleado_id = e.empleado_id
    INNER JOIN departamentos d ON e.departamento_id = d.departamento_id
    WHERE p.estado <> 'cancelado'
    GROUP BY d.region_id, e.empleado_id, e.nombre, e.apellido
),
ranking_region AS (
    SELECT
        region_id,
        empleado_id,
        vendedor,
        total_ventas,
        ROW_NUMBER() OVER (
            PARTITION BY region_id
            ORDER BY total_ventas DESC
        ) AS rn
    FROM ventas_empleados
)
SELECT
    r.nombre AS region,
    rr.vendedor,
    rr.total_ventas
FROM ranking_region rr
INNER JOIN regiones r ON rr.region_id = r.region_id
WHERE rr.rn = 1
ORDER BY rr.total_ventas DESC;
```

regiones 1

WITH ventas_empleados AS (SELECT d.region_ | Enter a SQL expression to filter results (

Grilla | Texto

	region	vendedor	total_ventas
1	Norte	Ana Ramos Cruz	4.032.536,21
2	Este	Eduardo Moreno Ramos	3.620.249,93
3	Centro	Mónica Martínez Reyes	3.174.764,32
4	Sur	Patricia Castro Díaz	2.790.455,11
5	Oeste	José Jiménez Hernández	2.737.425,23

```sql
--EJERCICIO 5--

SELECT
    c.cliente_id,
    c.nombre || ' ' || c.apellido AS cliente,
    CURRENT_DATE - MAX(p.fecha_pedido) AS recency_dias,
    COUNT(p.pedido_id) AS frequency_pedidos,
    SUM(p.total) AS monetary_total
FROM clientes c
INNER JOIN pedidos p ON c.cliente_id = p.cliente_id
WHERE p.estado <> 'cancelado'
GROUP BY c.cliente_id, c.nombre, c.apellido
ORDER BY monetary_total DESC;
```

clientes 1

SELECT c.cliente_id, c.nombre || ' ' || c.apellido A | Enter a SQL expression to filter results (use Ctrl+Space)

Grilla | Texto | Record

	cliente_id	cliente	recency_dias	frequency_pedidos	monetary_total
1	1.614	Ramón Ramírez Sánchez	189	4	1.268.642,43
2	944	Alejandra Jiménez Gómez	269	5	1.018.382,79
3	1.561	Ramón López Vargas	291	5	941.276,81
4	1.597	Fernando García Mendoza	255	6	912.110,66
5	1.189	Andrés Rivera Díaz	150	4	882.033,34
6	1.516	Eduardo Cruz Vargas	131	5	872.831,55
7	365	Gabriela Castro Moreno	180	6	871.608,5
8	1.595	Daniela Rivera Torres	560	4	868.892,89
9	900	Claudia Cruz Pérez	333	6	847.695,03
10	613	Isabel Martínez Díaz	160	6	841.196,7
11	1.892	Manuel Sánchez Gutiérrez	604	5	836.287,69
12	322	Elena Gómez Sánchez	735	3	826.859,95
13	1.958	Ramón Pérez Aguilar	351	2	773.125,56
14	931	Isabel López Reyes	238	5	765.586,67
15	1.046	Valeria García Castro	241	3	757.327,06
16	1.450	José López Núñez	135	5	756.612,24
17	796	Diego Flores Herrera	140	9	746.330,98
18	794	Francisco Herrera Moreno	324	4	746.139,29
19	1.287	Sergio Díaz Ortega	168	6	740.593,32
20	655	Mario Jiménez Mendoza	404	6	740.216,05
21	1.154	Sergio Martínez Mendoza	160	4	735.911,29
22	1.660	Juan Torres Rodríguez	318	4	735.318,59